

Guangxun Zhang

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EDUCATION

New York University

Sep 2023 - May 2027 (expected)

B.A. Honors Mathematics and B.A. Honors Computer Science

New York, NY

Mathematics GPA 3.93 / 4.00 | Computer Science GPA 3.87 / 4.00

AWARDS AND FUNDING

Computer Science Prize for Academic Excellence

2025 - 2026

Awarded to one student per year by the Department of Computer Science in recognition of academic excellence.

Presidential Honors Scholar

2024, 2025, 2026

Awarded for ranking within the top 10% of the entering first-year class.

Summer Undergraduate Research Experience Grant \$4,400

2026

Selected as one of approximately 12-14 students annually for a competitive interdisciplinary mathematics research program.

Dean's Undergraduate Research Fund Grant \$2,000

2025

Awarded for demonstrating the skills required to conduct an independent research project.

PUBLICATIONS

- [1] **Guangxun Zhang**, Mason Haberle, and Davi Geiger. "Stable Mean Flow: Lyapunov-Inspired One-Step Flow Matching." *CVPR 2026. Highlight paper (top 2.8% of all submissions).* [Paper]
- [2] Zichen Jeff Cui, Omar Rayyan, Haritheja Etukuru, ..., **Guangxun Zhang**, ..., Lerrel Pinto, and Nur Muhammad Mahi Shafiullah. "Contact-Anchored Policies: Contact Conditioning Creates Strong Robot Utility Models." *Robotics: Science and Systems 2026.* [Project]
- [3] **Guangxun Zhang** and Xiaomei Wang. "Multi-Scale Pyramid Feature Integration and Style-Mixing Contrastive Analysis for Robust Invasive Ductal Carcinoma Detection." *NTCI 2025.* [Paper]
- [4] Xintong Zeng, Xiaomei Wang, and **Guangxun Zhang**. "Comparison of Different Models for Natural Gas Forecasting and Assessment." *ICDLT 2025.* [Paper]

MANUSCRIPTS

- [5] **Guangxun Zhang*** and Brian Cai*, Boxuan Zhang, Chao Chen, and Ruixiang Tang. "JIVE: Jacobian-Informed Volume Expansion for Diverse Generative Sampling." *Under review at ICLR 2027.* [Manuscript] * Equal contribution.
- [6] Xuanzheng He, **Guangxun Zhang**, and Davi Geiger. "A Dataset Is More Than a Bag of Pairs: Preserving Same-Scene Acquisition Structure for Physical Degradation Learning." *Under review at WACV 2027.* [Manuscript]

RESEARCH EXPERIENCE

JIVE Diverse Generative Sampling

May 2026 - Sep 2026

TRAIL Lab, Rutgers University | Research Intern | Advised by Prof. Ruixiang Tang

Co-developed JIVE, a Jacobian-informed latent-branching method for diverse diffusion sampling. Formulated a local volume-expansion objective and designed branching experiments to broaden samples while preserving prompt alignment.

Chain-of-Thought Post-Training

Nov 2025 - Present

Reasoning with AI Systems Lab, NYU | Research Assistant | Supervised by Prof. Pavel Izmailov

Built a compute-efficient pipeline for adding reasoning supervision to chess language models using Stockfish-selected tactical positions. Generated reasoning traces with a lightweight fine-tuned model and compared reasoning supervision with additional chess data under matched compute.

Learning Stochastic Dynamics from Snapshot Data

Nov 2025 - Sep 2026

Summer Undergraduate Research Experience, NYU | Research Assistant | Supervised by Prof. Wenzheng Shi

Developed a two-stage method for learning drift and diffusion from unpaired snapshot data, connecting it with the drifting model. Combined Euler-Maruyama transition likelihood with multi-lag conditional velocity matching and grouped displacement estimates to reduce Brownian noise.

Contact-Anchored Policies

Apr 2025 - Feb 2026

Generalizable Robotics and AI Lab, NYU | Research Assistant | Supervised by Prof. Lerrel Pinto

Built a SAM2-based pipeline to segment manipulated objects and generate contact-anchor labels for robot demonstrations. Supported zero-shot evaluations across environments, tasks, and robot embodiments.

Stable Mean Flow

May 2025 - Oct 2025

Courant Institute of Mathematical Sciences, NYU | Researcher | Advised by Mason Haberle and Prof. Davi Geiger

Developed Lyapunov-inspired regularization for one-step flow matching to control prediction-error growth along learned transport paths. Derived trajectory-drift bounds and built ablations to study sample quality, convergence, and stability.

Deep Learning for Invasive Ductal Carcinoma Detection

Jun 2024 - Jan 2025

University of Electronic Science and Technology of China | Research Assistant | Advised by Prof. Xiaomei Wang

Designed a multi-scale histopathology classifier combining cellular morphology with tissue context. Added a style-mixing contrastive objective for robustness to staining variation and led ablations for the resulting NTCI 2025 paper.

TEACHING MENTORING AND SERVICE

Undergraduate Research Mentor

May 2026 - Sep 2026

Image Dehazing with Generative Models | Kooick Industry Project, NYU

Mentored a student project on diffusion-based image dehazing.

Peer Mentor

Sep 2025 - Jan 2026

Mathematics and Computer Science | Peer Mentoring Program, NYU

Supported students with coursework, study strategies, and course planning.

Mathematics Grader

Sep 2024 - Present

Courant Institute of Mathematical Sciences, NYU

Graded assignments and exams in Calculus III, Linear Algebra, Numerical Analysis, and Honors Numerical Analysis.

Reviewer, International Conference on Learning Representations

2027

TECHNICAL SKILLS

Programming

Python, C/C++, Java, MATLAB, SQL, x86 Assembly

Machine Learning

PyTorch, JAX, TensorFlow, Hugging Face Transformers, Diffusers, scikit-learn, NumPy, SciPy

Systems and Tools

Linux, CUDA, Docker, Slurm, Git, Jupyter, Weights & Biases, LaTeX

Mathematics

Probability, Statistics, Linear and Abstract Algebra, Discrete Math, Real Analysis, Numerical and Convex Analysis, Calculus on Manifolds, ODEs, PDEs, SDEs, Measure Theory